Copilot

Certainly! The **Die Hard problem** is a classic puzzle that involves two water jugs of different capacities and the goal of measuring a specific amount of water using these jugs. Let's implement a solution in Java:

```java
import java.util.Scanner;

public class DieHardProblem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int T = sc.nextInt(); // Number of test cases

        for (int tc = 0; tc < T; tc++) {
            int a = sc.nextInt(); // Jug 1 capacity
            int b = sc.nextInt(); // Jug 2 capacity
            int c = sc.nextInt(); // Desired amount of water

            System.out.println(solve(a, b, c));
        }

        sc.close();
    }

    static String solve(int a, int b, int c) {
        // Check if it's possible to measure 'c' liters using the given jugs
        // (c <= max(a, b)) and (c % gcd(a, b) == 0)
        return (c <= Math.max(a, b) && c % gcd(a, b) == 0) ? "YES" : "NO";
    }

    static int gcd(int x, int y) {
        return (y == 0) ? x : gcd(y, x % y);
    }
}
```

Explanation:

- We read the number of test cases (`T`) and then for each test case, we read the capacities of the two jugs (`a` and `b`) and the desired amount of water (`c`).
- The `solve` method checks if it's possible to measure exactly `c` liters using the given jugs.
- We use the greatest common divisor (GCD) function to check divisibility.

Compile and run this code, and it will determine whether it's possible to measure the desired amount of water using the provided jugs. If possible, it prints "YES"; otherwise, it prints "NO". 🚰🔍